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Quantum Ensemble Models for Healthcare and Life Science

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Quantum Ensembling Methods for Healthcare and Life Science

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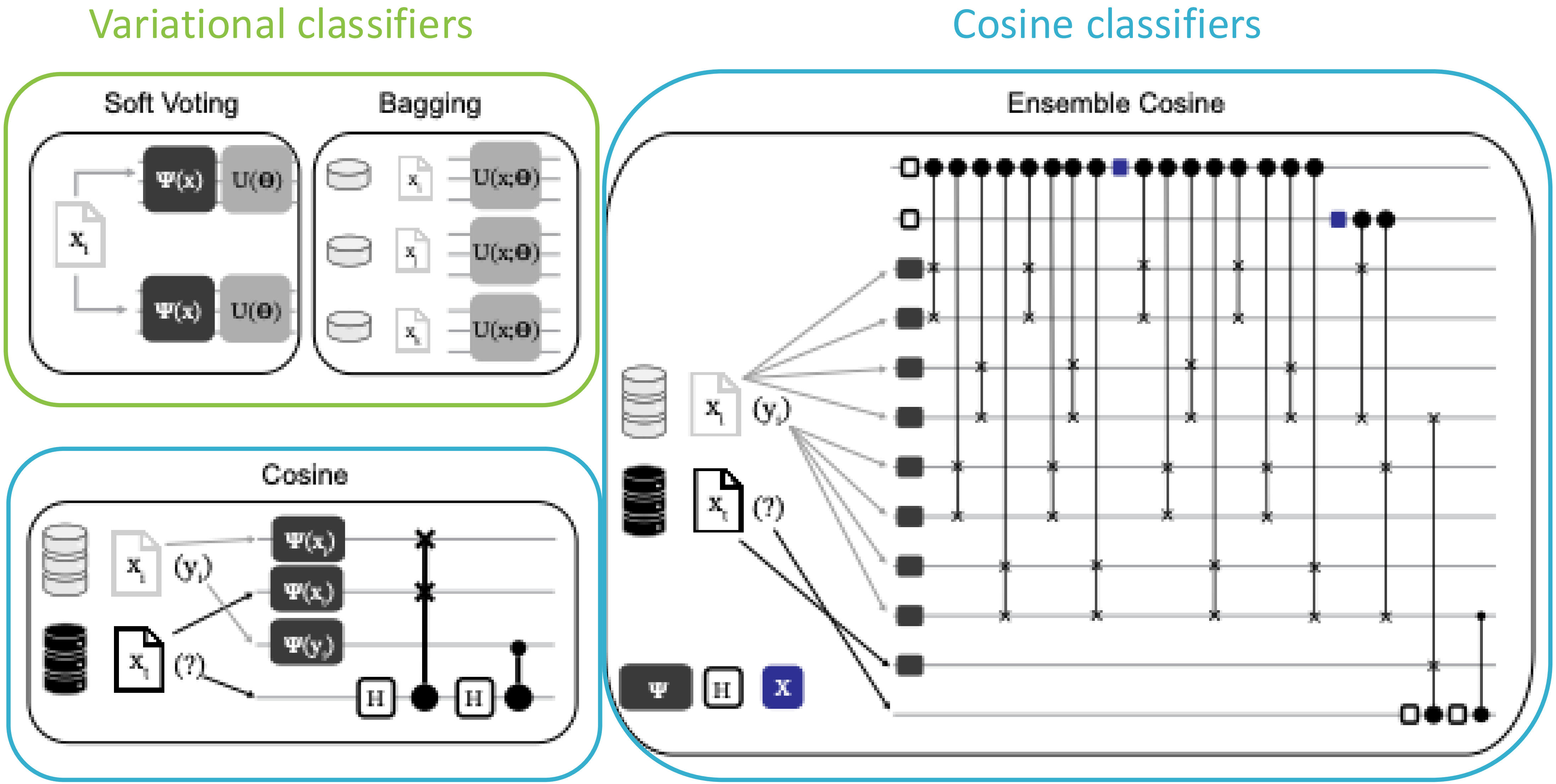
- We completed an empirical study that compares the performance of different quantum ensembling approaches when trained on HCLS data
- Why ensembles?
  - QML models run on hardware often faces bottlenecks from number of circuits, circuit depth, and measurements
  - Ensemble methods can potentially alleviate these bottlenecks by training a collection of weak learners that can be implemented with smaller, shallower unitary circuits.

- Our contributions are:
  - Performance comparison of different aggregation methods
    - classical (boosting, bagging and soft-voting)
    - quantum (bagging via superposition and perturbation)
  - Overhead comparison for different ensembling workflows:
    - serial processing using variational learners
    - parallel processing using cosine learners
- Demonstration of efficacy in real-world HCLS data taken from renal cell carcinoma (RCC) patients treated with immunotherapy
- Tested circuits from 7-56 qubits, including experiments on QPU
- Submitted to IEEE Quantum Week 2025

Tested ensemble classifiers

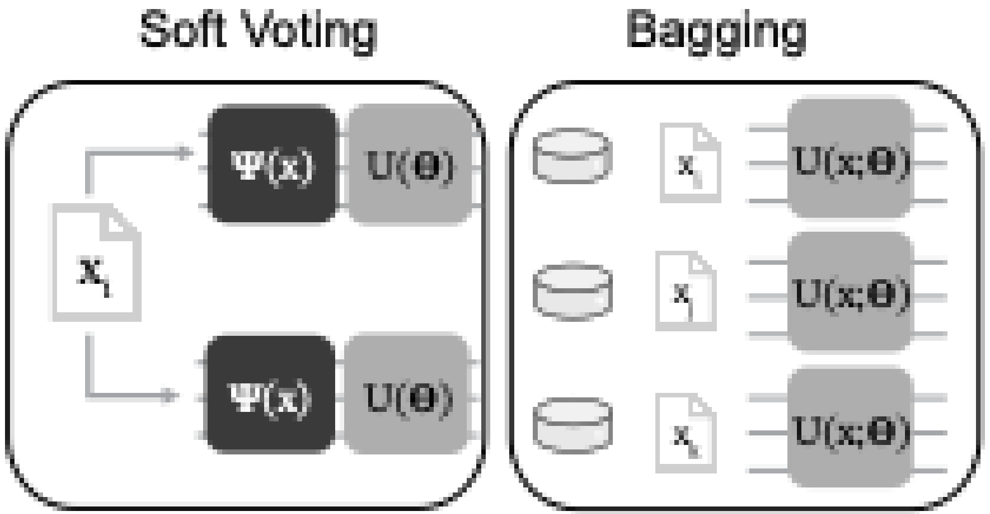


Variational classifiers

Each learner predicts the label of one encoded feature vector x_i at a time using a parameterized unitary

Soft Voting Ensemble

- An ensemble of parameterized circuits are trained using batch gradient descent (cross entropy loss)
- One sample is embedded and each wire (learner) makes an independent prediction

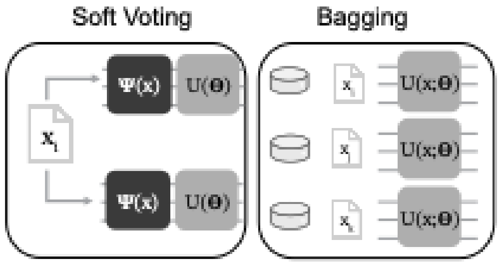


Bagging Ensemble

- An ensemble is composed from multiple learners (parameterized circuits) each are trained on a different subset of the data
- One learner is trained at a time then during inference each learner makes a prediction on an unknown sample

- With (n) qubits per learner, (L) learners per ensemble and (p) layers in each parameterized circuit the soft voting ensemble has a total of $(3(L+1)n\ell)$ trainable parameters
- With (n) qubits per learner, and (p) layers in each circuit, the bagging ensemble has a total of $(3(L+1)n\ell)$ trainable parameters but only $3(L+1)$ are trained at a time

Variational classifiers

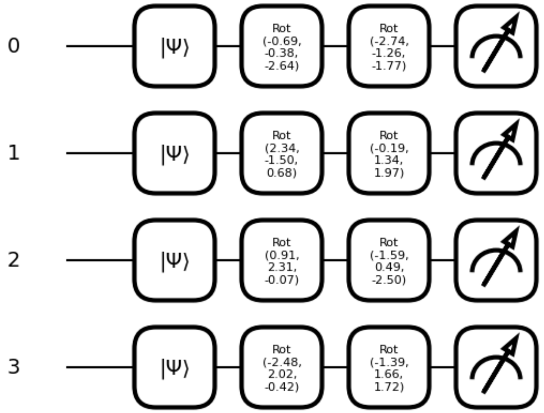


Boosted classifier

- We tested a 3rd variational classifier that used AdaBoost
- AdaBoost iteratively updates the weight of samples in the training set based on the error of the weak learner

Configurations

- Amplitude encoding of features
- A parameterized single qubit rotation decomposed as a RZ-RY-RZ gate sequence is applied to each qubit
- If learner has 2+ qubits, a CNOT is applied between qubits
- Another parameterize rotation is applied to each qubit
- Trained using mini batch with Adam

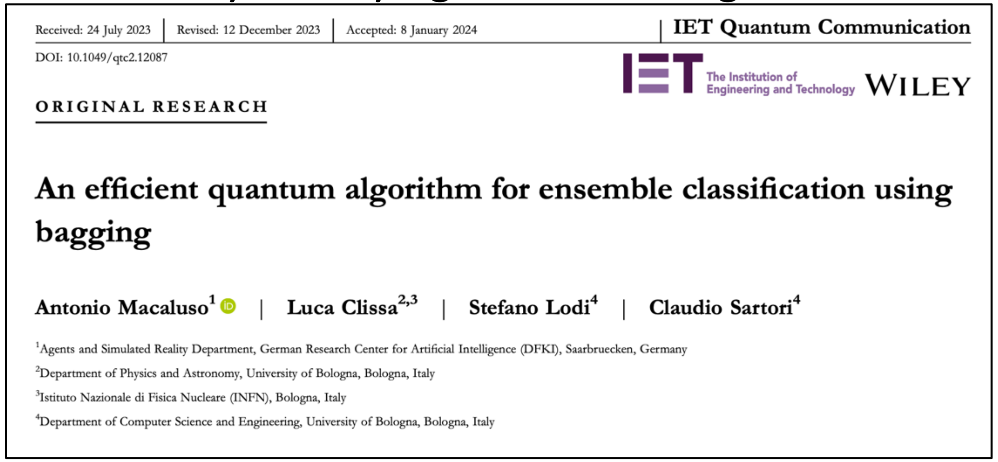


90 configurations tested over the following parameters:

- Adam learning rates $\alpha \in [1 \times 10{-}3, 1 \times 10{-}2, 1 \times 10{-}1]$
- Batch size $b \in [1, 2, 4, 8, 16]$
- Ensemble sizes $n_\ell \in [1, 2, 3, 4, 5, 6, 7]$

Cosine classifiers

- Predicts using multiple encoded feature vectors x_i, x_t and known label information y_i
- Quantum cosine classifier (QCC) is a weak learner that calculates the cosine distance between sample vectors in quantum state via interference using a single training sample
- Quantum ensemble cosine classifier (QEC) captures independent trajectories via sampling in superposition from 2^d transformations of x_i where d is ancilla control register qubits → exponential increase in ensemble size with linear increase in circuit depth
- Implemented by modifying and extending Macaluso et al (2024)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ORIGINAL RESEARCH

An efficient quantum algorithm for ensemble classification using bagging

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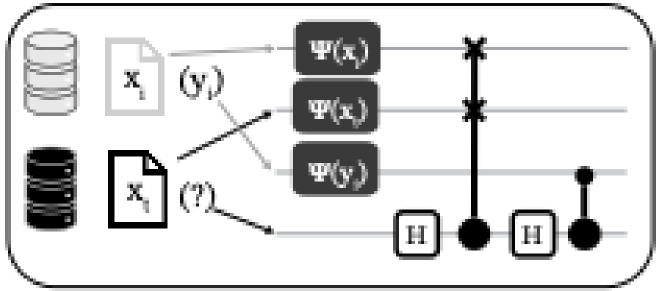
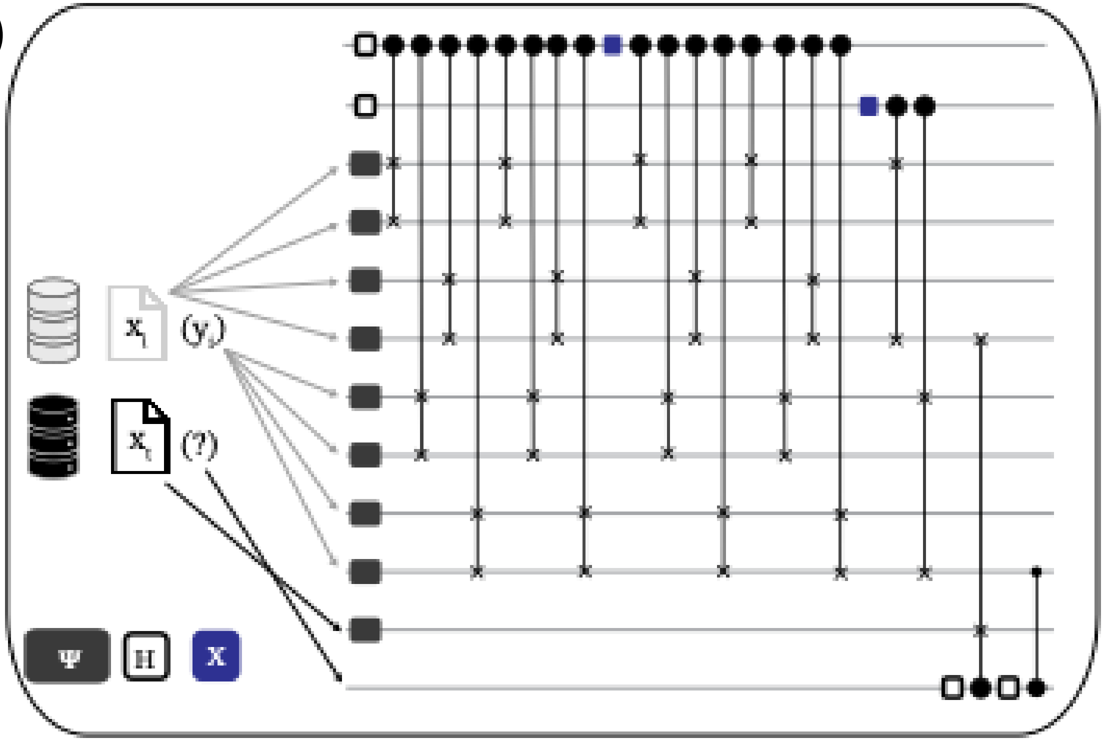
Cosine

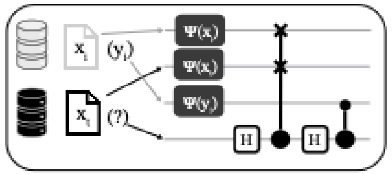


Ensemble Cosine

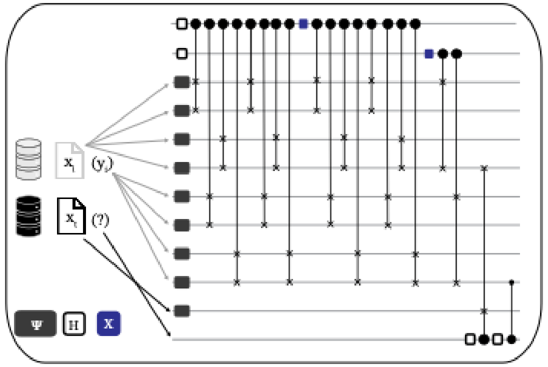


Cosine classifiers

Cosine



Ensemble Cosine



QEC with Random Unitaries (QECRU)

- QEC's structure yields unitaries that are generally uncorrelated
- We tested a 3rd quantum cosine classifier that considers other forms random sampling on *U(n)* to make all distribution on *U(n)* as a possible choice in the sampling, in this study used a uniform distribution on *U(n)*

Simulation configurations tested over the following parameters:

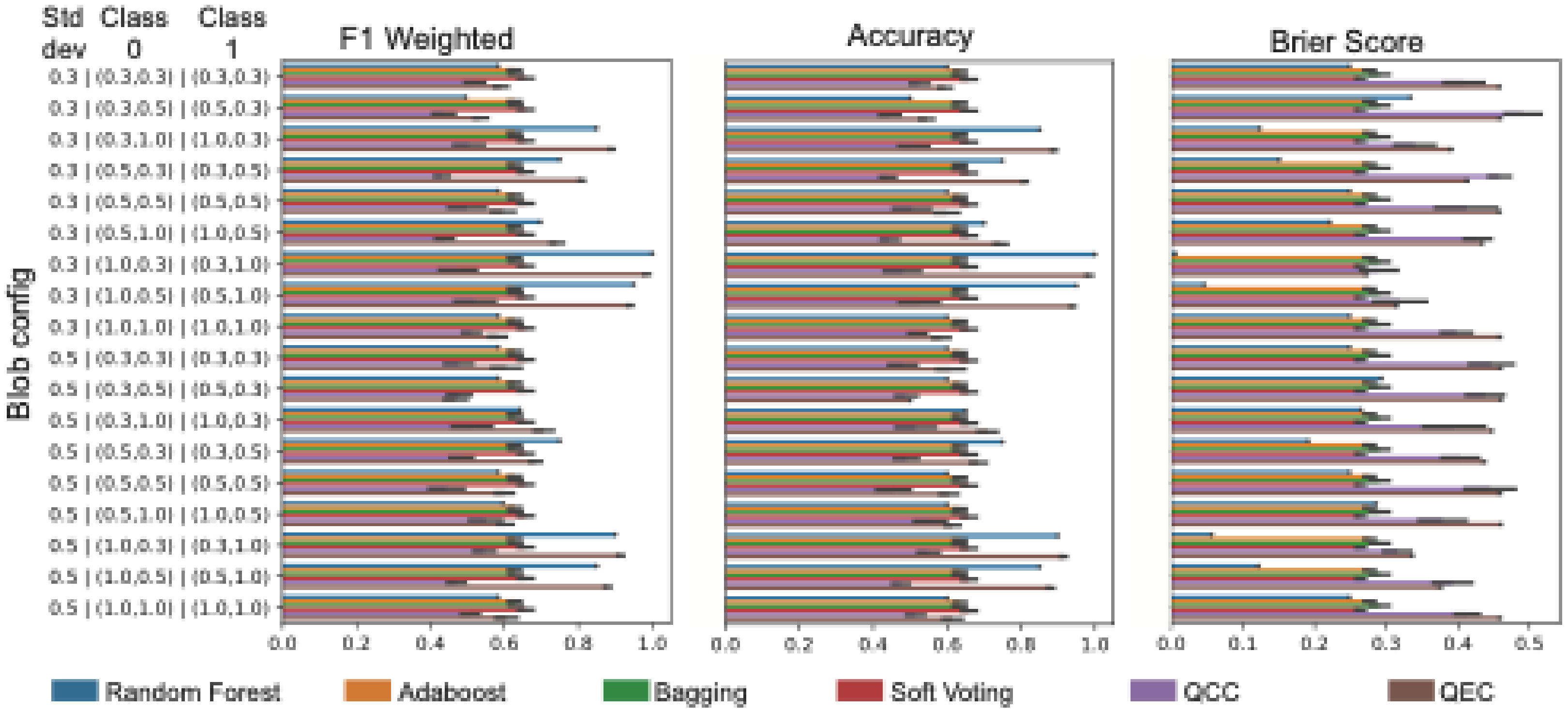
- Control registers $d \in [1,2,3]$
- Training size $n\_train \in [2,4]$
- Number of random swaps $n\_swap \in [1,2,4]$
- Number of features $n\_feature \in [2,4,8]$

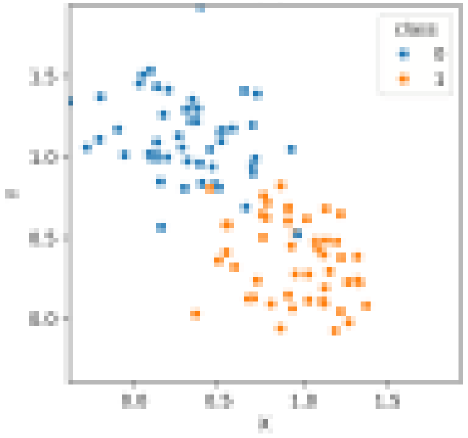
Testing on synthetic data

Generated 18 different configurations of gaussian blobs over all combinations of the following parameters:

- *cluster std = [0.3, 0.5]*
- *p1 = [0.3, 0.5, 1.0]*
- *p2 = [0.3, 0.5, 1.0]*

Where *p1* and *p2* represent the x and y coordinates of the centers of class 1 (*p1*, *p2*) and class 2 (*p2*, *p1*) blobs

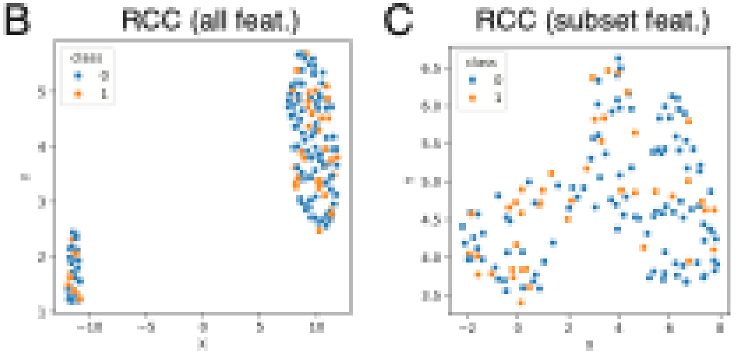


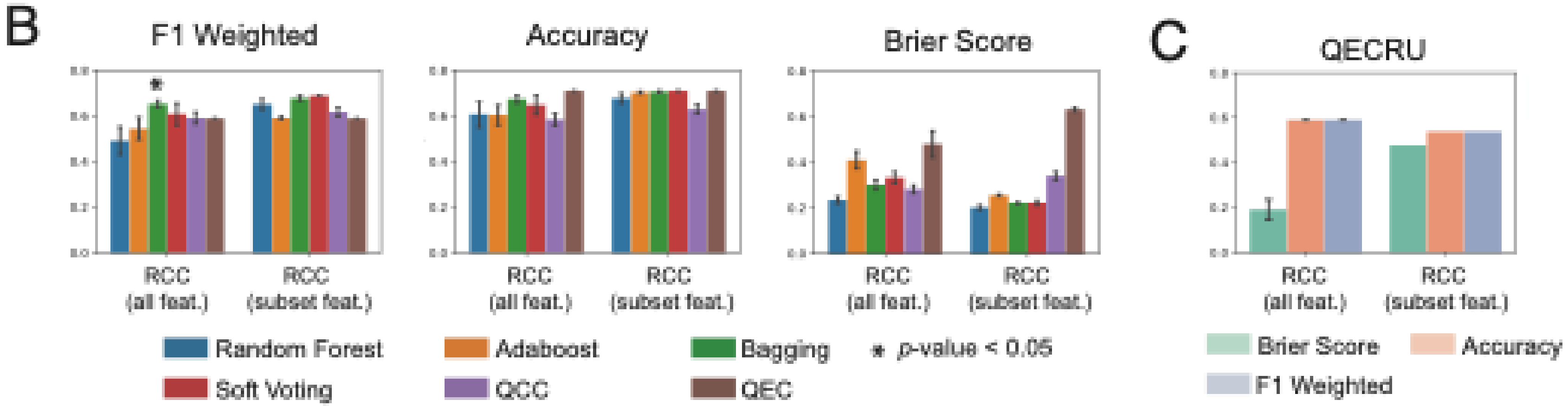


- Variational approaches slightly outperformed the RF and cosine classifiers for many of the blob configurations
- Blob configurations whose centers were further apart saw the RF and QEC outperforming other approaches
- It may be the variational methods are relatively over-parameterized for these simplistic two feature datasets.

Testing on Renal Cell Carcinoma

- Used DESeq2 with variance stabilizing transformation (VST) to normalize RNA-seq mRNA counts for 150 patient samples from McDermott et al
- Class labels were response to immunotherapy
- Created subset of the feature space by selecting 8 genes known to be associated with immunotherapy response (*CD8A, CXCL9, CXCL13, IFNG, CD274, PDCD1, VHL, GZMK*)
- 10 80/20 train test splits
- PCA for dimensional reduction and PCs used as features





- Variational methods performing well with the bagging classifier significantly improving on the RF
- The other quantum classifiers were doing well compared to RF with a modest, though insignificant, increase of the F1 or accuracy by ~10%.
- While the performance of QEC and QECRU was not significantly improved over the variational and RF classifiers, it was able to achieve its promising performance only using anywhere from 2-4 training samples
  - This suggests the potential for significant applications to cases where the amount of data is highly constrained

Testing on Renal Cell Carcinoma

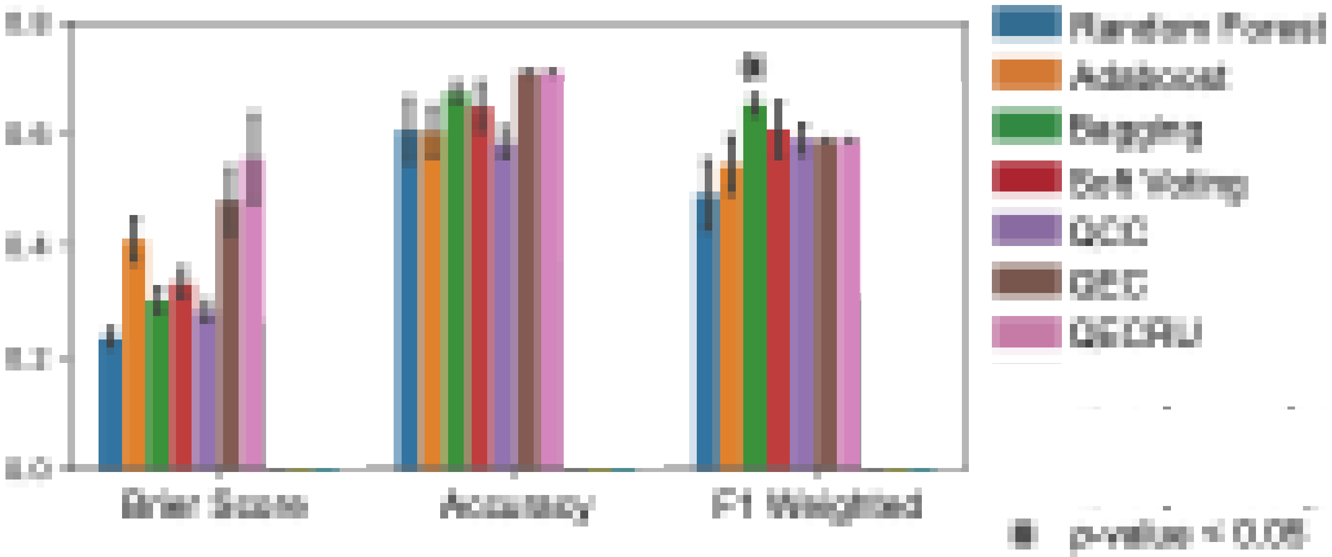
- We find that for both the variational and cosine classifiers the qubit number grows logarithmically with features
- Cosine qubit size grows linearly with training size
- Cosine classifier depth increases linearly with exponential increase in ensemble size

Model	Features	Qubits	Depth	2Q depth	F_1
Single Classifiers					
Cosine	2	4	64	11	0.77
Variational	8	3	59	14	0.78
Ensemble Models					
Cosine	4	16	132	27	0.72
Random Unitary	8	8	143	30	0.66
Soft Vote	4	10	22	12	0.78
Bagging	8	9	57	14	0.81
AdaBoost	2	6	6	0	0.68

TABLE I

COMPARING BEST PERFORMING ENSEMBLES ON AN INDIVIDUAL RCC TEST DATASET SPLIT USING A PCA FEATURE EMBEDDING FROM THE FULL FEATURE SPACE, AND REPORTING ON CIRCUIT WIDTH, CIRCUIT DEPTH INCLUDING MEASUREMENT, FEATURE SIZE, AND WEIGHTED F_1 SCORE.

Testing on Renal Cell Carcinoma using QPUs



- Tested QEC on 127 qubit Eagle R3 device over five splits using 8192 shots and Pauli twirling (PT) and dynamical decoupling (DD)
- 7 qubit QEC was configured as: d=1, n_train=2, n_swap=1, and n_feature=2
  - overall depth = 100; 2-qubit depth = 20
  - underperformed other ensemble classifiers and RF

- 56 qubit circuit first with PT and PT+DD configured as: d=2, n_train=8, n_swap=1, and n_feature=32
  - overall depth = 853; 2-qubit depth = 201
  - PT + DD improved over PT only and F1 was comparable to other cosine classifiers and improved over RF in noise-free simulation
  - Much lower Brier scores that QEC and QECRU

Other findings

Variational methods could learner with fewer samples if the number of learners is increased

Bagged and boosted ensembles can be simulated quite fast as a learners are trained one at a time on disjoint subsets

Boosted ensembles can't be easily distributed because of the need to adaptive weighting of samples

Soft voting is the slowest with the highest memory requirements as the gradient update step is expensive

- There can be opportunities to better optimize the gradient update

Conclusions and future work

Quantum ensemble classifier combine the strengths of classical ensemble learning and quantum computing

Quantum ensembles were able to reach comparable performance to RF with orders of magnitude fewer training samples

QEC as a non-variational ensemble may have some advantages in scaling on hardware

We did see performance positively trending with additional error mitigation methods

- We should test on newer Heron devices and with increased circuit complexity

We are exploring alternate quantum ensembles looking at quantum optimization techniques

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